

ActInf GuestStream #015.1: Bobby Azarian, Universal Bayesianism: A New Kind of Theory of Everything

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yep all right thanks daniel for having me everyone at the active inference lab uh sorry for canceling twice uh life's been kind of crazy with uh finishing uh this book that this talk will be about and having a kid i'm six months old and i i got sick like a week ago like right leading up to like you know didn't want to cancel a third time and look crazy so this is a very rushed presentation um it'll be better if it's more of like a conversation since i haven't even gone through the slides myself once um and it's gonna get into a lot of things like immediately like open up bags of worms like there's just philosophical issues um things that people won't agree about like immediately but um yeah so this is just going to be like a small introduction to the thesis in the book um and yeah i'll explain more i guess uh after i begin so yeah it's uh call this the integrated evolutionary synthesis um compared to like the modern synthesis or even the extended evolutionary synthesis this is integrated because it uh kind of reconceptualizes evolution uh biology and evolution in terms of the in terms of thermodynamics and information theory or energy flows energy and information flows um takes from cybernetics so yeah it's a non-reductive theory of everything that includes emergent phenomena like life mind and civilization um i say that because most theories of everything we're used to are fundamental physics theories that don't have anything to say about life and consciousness they ignore them um and uh i don't think those should be called theories of everything because theories of everything should be uh an explanation for everything and if you leave the things that we care about most out then it's not much of a theory of everything so this is the book i've spent the last two to three years riding obsessively and uh it's available for pre-order now so if you want to check that out um you can find it with a google search uh here's the table of contents um so the book is broken into three sections uh part one uh talks about the origins of life and part two about evolution while part three focuses on consciousness and free will and the fate of life in the universe so these like really big questions but we're gonna focus on part two and uh give a pretty like a superficial overview of part two um but you can feel free to ask

me uh any questions about uh any of the other topics that are related and i'll try to see what i can answer without going into like all of the information i would i would need to explain those things um so thomas huxley known as darwin's bulldog because he was such a strong advocate for darwin's theory said that the question of questions for mankind the problem which underlies all others is more deeply interesting than any other is the ascertainment and of the place which man occupies in nature and of his relations to the universe of things so where do we fit into the grand cosmic scheme um most scientists in the 20th century and i would even say now um believe uh that we are an accident so uh jacques mano french biochemist who won the nobel prize for uh physiology and medicine um wrote a book called chance of necessity which was really influential and he was just as reductionist as you can get man at last knows that he is alone in the unfeeling immensity of the universe out of which he emerged only by chance neither so some of the words are covered by uh by daniel so neither is his destiny spelled out or his duty have been written down i think there's something wrong with that quote but um so if there are times where i'm trying to read something where the picture's over i might have to skip you can you can move that because it looks on my side it it looks fine perfect thank you great yeah um neither his destiny or his duty have been written down so that was kind of based on this old idea that the emergence of life uh was a product of chance assembly so like this uh just a statistical fluctuation that brought together all the molecules needed to create the first cell we now know that that's probably not what happened it's this gradual process of self-organization jeremy england has uh been in the media a lot about his theory of dissipative adaptation so um let's so we will see uh this is kind of like a different point of view not everybody was convinced by a mono uh carl sagan said the origin of life must be a highly probable affair as soon as conditions permit up it pops um so this is an important question um whether life was uh improbable or inevitable so if life is highly improbable we're likely to be alone in the universe but if the emergence of life is inevitable given a certain set of common uh physical conditions then it's likely that we have life um at least on planets that are sufficiently earth-like uh specifically have uh geochemistry like earth's uh uh have star uh some uh like the earth's because that's what uh provides the energy that pushes it far from equilibrium so we're going to get in to all that stuff pretty soon um christian to do this quote is kind of a response to stephen j Gould who just really emphasized uh to both mono and stephen jay Gould the importance of chance uh in nature and that everything is basically you know a contingency that came from a chance process um where did says that you know you can have this inevitability that some philosophers would see as as teleology but

it doesn't mean that uh this inevitable progress is being driven by the supernatural force he explains that the natural constraints within which chance operates are such that evolution in the direction of increasing complexity was virtually bound to take place if given the opportunity chance does not exclude inevitability so he was also a nobel prize winning uh biologist the same category as mono but had a complete opposite opinion and we can talk about that too basically when uh systems are open systems and they're pushed far from equilibrium by a flow of energy you basically uh get something like a statistical bias away from what we would expect with isolated systems closed systems that aren't uh open to energy will relax the thermodynamic equilibrium and when a system is being pushed far from equilibrium uh it will naturally organize because uh organization happens uh under the flow of energy and we're gonna see exactly why it's a darwinian process but basically um you get inevitability if you have the right ingredients we'll talk about to kind of frame this we already kind of mentioned those two world views so the first world view would be the reductionist world view and i should be clear that um productionism is uh also it's it's a method and an ideology and we shouldn't uh confuse those two things um and daniel i guess just so i'm you know know where i'm at maybe when you when i've gone 30 minutes you can let me know okay i'll hoot okay great um so uh reductionism the method uh basically says that reality can be understood by breaking down all physical phenomena into their simplest parts so that we may observe the basic behavior of the fundamental constituents of nature so i'm just going to plug in this charger i might need to do that a little bit but um basically uh the method is how science works if we want to understand emergence we also have to understand how the components that systems are made of uh work in isolation it's an essential part of the story but it's not the whole part we'll see um so the method inspired an ideology and ideology is is just that it's it's an ideology and it's rigid and um basically uh the method uh doesn't really imply that the ideology is true so um when i'm attacking reductionism i'm not attacking the method i'm i'm attacking the worldview and so that view we can say that um hard determinism is is part of the reductionist worldview so uh thought experiment that some people are familiar with i'm sure la place is demon but basically uh laplace was a french mathematician that uh took uh newton's kind of model of reality uh and applied it to everything newton himself was kind of a mystic when it comes to life he didn't really think that physics applied to to organisms or at least to humans but laplace really you know came up with the idea that like uh basically the evolution of the universe is just this like clockwork universe where it's like um a machine with cogs that are just turning so there are no uh there's no room for agency and free will in that

picture when boltzmann came along uh in the 19th century and came up with statistical mechanics the clockwork universe kind of became the billiard ball universe but it's just the idea that everything in the universe is made of atoms and that these atoms are um interacting and that's the whole picture so the ideology says we're nothing more than our atoms so we don't make decisions uh consciousness is just an epic phenomenon and it feels to us like we are but actually this trajectory was determined basically uh not basically from the moment of the big bang and there's kind of no freedom in anything and uh no future other than what was determined from the beginning the second law of thermodynamics also sort of shaped the reduction of soil abuse as the universe is becoming increasingly disordered there's actually kind of two different interpretations of the second law uh the first one kind of emerged from the work of uh carnell and clausius the kind of original thermodynamics work where entropy was about um a measure of uh energy that can't be used for work so a kind of measure of like useless energy um where later with boltzmann he tried to put the second law on like a statistical basis based on like atomic theory so basically uh people applied uh boltzmann's statistical interpretation of the second law to the universe and came up with this notion that the universe is becoming increasingly disordered which isn't great for life because it means that life is transient and likely cosmically insignificant and this worldview is associated with materialism uh which doesn't recognize consciousness is real certainly doesn't recognize it as having causal power and historically it often ignored apparently a material phenomenon like energy and information and mental processes i am just going to grab my charger so sounds good this doesn't die but i will be right back take your time those who are watching live please feel free to leave a comment or a question i'm writing some things down we'll look forward to the return of electrical power to bobby all right my first live stream i will be more prepared when i return to part two that's something else i've told daniel is that this is kind of like part one and i like to come back and talk about the ideas that come after it will have 15.2 it'll be awesome it's an infinite sequence it's the sequence is always yours great okay um so reductionist worldview says that all life forms including humans are nothing more than collections of atoms obligatory following fixed and arbitrary mechanical trajectories determined solely by math and not by mine this world view if if you convince someone that they don't have free will because you tell them about the reductionist worldview um there was actually a study where they did this and they read participants a passage from francis crick explaining that uh it appears that we don't have any free will um the participants actually on subsequent tests uh were more like more likely to cheat on the test because they felt like they

didn't have any personal responsibility they weren't making the decisions therefore they could cheat that's kind of funny because they are actually you can see that like by what they're like being told they are changing uh their behavior so it is having some kind of effect if we do in fact have an agency or free will and we're kind of talking about agency uh we don't really get into free will but those are related topics but if that's not true then uh that's really bad that the majority of scientists are telling people we don't have free will because the people that believe this it's having negative effects on for example mental health other studies have shown that belief and no free will can lead to depression and belief in free will can have all these positive effects um of course soon as i start talking about free will i'm sure a lot of people have objections uh might think that there's no way that we can have room for free will in a physical world view um but that's exactly what we're going to begin to talk about today uh the causal power of information and that basically organisms are agents that are cybernetic control units so as controllers we do have agency and the belief that were accidents of nature rather than natural manifestations of physical laws uh i'm sure also uh impacts our sense of well-being just so you know i'm not making this up um i can i'll give you some quotes from really famous physicists who are uh kind of militant reductionist brian green says i think it's very important to face up to the truth of reality which is in fact that life and is a fleeting phenomenon on the entire cosmological timeline brian green just came out with a book and he's like going on interviews just saying this and it's really interesting because some other famous physicists don't agree at all and he's stating it like it's a complete fact so for example david deutsch would not agree with this and we'll get into that um sabine hostenfelder is another person um i know some of the some of my friends in the community um who are these people you know working on emergence and causation and stuff uh get bothered by her statements because she's also very uh ultra reductionistic so i wish people would stop insisting they have free will it's terribly annoying insisting that free will exist as bad science like insisting that horoscopes tell you something about the future it's not compatible with our knowledge about nature so uh i'm saying that's wrong we won't get into free will that much but if you have questions at the end we will talk about how information adaptive information uh gets built up by evolution in uh adaptive systems complex adaptive systems and you will see how uh these systems actually do control their own future so sam harris is another anti-freewheel person says you're not controlling the storm and you are not lost in it you are the storm so it's kind of funny because he's using a metaphor here but what this story kind of reveals is that uh yes we are a storm that's right we are a dissipative structure like a storm uh emerges to dissipate uh an energy

gradient and uh we're not really gonna talk about that that's what the first part of the book is about the origins part but um i'm just pointing it out because it's interesting because we are like a storm in that aspect but we also differ from a storm because a storm does not have agency a storm does not have the ability to seek out new energy gradients it's just kind of being pushed around by these gradients in nature where uh organisms are actually controllers and so you are storm but you do have control over the storm these people are kind of have the opposite view so julio tinoni uh creator of integrated information theory says there is true free will a great lecture that just popped up i think months ago where he gives a two-hour talk on uh basically integrate information and uh how it shows that systems do have causal power and uh in a way uh have free will so depending on your definition of free will we do have free will according to his model david deutsch said in an interview with john horgan just uh a year ago or so i'm sure we have free will christoph koch says explains how you know there are a lot of experiments the lebet experiments which we will talk about next time that uh basically yeah some people interpret it as as not having free will but actually uh we will see later um when you deliberate uh that's actually an act of free will these choices were actually where you bring your entire conscious being to that question and try to analyze it under all the various conditions and actually uh there was a study coke did that kind of showed that the lebate lebet study uh um was kind of interpreted wrong and um for anyone who understands that stuff uh the readiness potential that was supposed to be proof that there was no free will disappeared when the decisions uh involved uh deliberation so yeah that's kind of meandering but i'm sure a lot of people are interested in this bigger picture of why this emergence paradigm is so radically different from the reductionist paradigm it kind of uh puts information front and center and systems that process information uh complex adaptive systems are computational systems and these systems uh in the with these systems information actually does have causal power we're gonna explain what we mean by that so paradigm of emergence uh the collective says the collective behavior of interacting parts not simply how they function in isolation is key to understanding the emergence and evolution of all the fascinating organisms and ecosystems that make up the biosphere so it's really not about just like particles it's about the patterns that emerge when uh these systems find these certain configurations uh through basically darwinian mechanisms and we'll see that self-organization too uh has a darwinian dynamic each emergence in the cosmic self-organization process is brought about by higher order phase transition that moves life ever further away from a state of thermodynamic equilibrium and total disorder i'm assuming the audience knows uh that what thermodynamic

equilibrium is it's basically uh if you have for example boltzmann took a model of um an ideal gas and explained that for example if the gas molecules are bunched up in the corner uh naturally it's going the gas is going to spread out until there's no pattern and uh basically it's like a completely mixed up state um so equilibrium is associated with death and disorder uh because systems naturally tend toward this more mixed up unpatterned completely chaotic state of course later we're gonna talk about uh why open systems that that rule doesn't apply basically systems that are open to the flow of energy can resist this tendency towards a disorder and that's really what the whole story is kind of based on um so through a nested series of such phase transitions these are also called these higher order phase transitions are major evolutionary transitions john maynard smith talked about them and meta system transitions by the cyberneticist valentine terchin basically the same concept of some subtle differences so the idea is that cosmic self-organization is a series of these transitions where functional things come together to make larger functional ones which come together to even make even larger ones and so on and through this process adaptive complexity or life so that's a more general way to talk about life becomes better equipped to dominate the cosmos life requires an organization that is increasingly hierarchical and integrated and therefore more resilient and computationally powerful here are just a couple of figures you see this hierarchy uh in this structural hierarchy of matter in life this was adapted from a big physics journal i can't remember which one it was but um basically you know physicists are recognizing this hierarchy even when it extends to life so jeffrey west's book scale comes to mind and so we see the hierarchy of science kind of naturally mirror this hierarchy of matter and life and um the reductionist picture would say that these things at the top here are kind of insignificant parts of the story transient but we will see that this you know the more we go up the more influence on cosmic evolution the phenomena have uh thomas huxley's question about a man's place in the cosmos uh is um basically the emergence of paradigm uh the paradigm of emergence i'm going to discuss and actually this is kind of maybe like in a specific or a new paradigm of emergence because it includes all this stuff about thermodynamics and information theory um says that emergence the emergence and evolution of life mind societies and technology are all part of one thermodynamic process one evolutionary process one computational process unified by the concept of knowledge uh this theory is a theory of knowledge creation and so knowledge is going to be the unifying theme and we'll briefly talk about epistemology i wanted to talk about it more but it just would have been too long um cosmic evolution uh in this paradigm is a universal process of becoming as opposed to being

so the universe isn't this static thing uh it's actually evolving very much like an organism or a complex adaptive system um so it's not a pan psychic theory in this theory consciousness emerges that's very important um but you might call it emergent pan-psychism uh because uh as this evolutionary process proceeds uh inanimate matter as life spreads gets converted into animate matter um so in this picture humans are neither a cosmic accident nor the end goal of so we are instead a step on the evolutionary ladder of becoming and we are also potentially an essential driver of increasing cosmic complexity so this big idea is that the universe is undergoing this majestic self-organizing process and at this moment of time at least in this corner of the universe we are the stars of the show um this picture is from eric chasen's uh book the rise of complexity uh cosmic evolution the rise of complexity he's a harvard astrophysicist but i just used freeman dyson code here because it seemed relevant it's conceivable that life may have a larger role to play than we have yet imagined life may succeed against all odds and molding the universe to its own purpose so i guess brian green doesn't buy that at all but um if that's true then uh as phys as cosmology cosmologists have to think about the role of life there's a lot of people who've said similar things like seth lloyd and famously raker as well um just a couple quotes here so you know you know when when people hear these terms progress there's just the stigma immediately they thought like oh someone's trying to sneak in religion or some intelligent design theory so you see these trusted names of physics here david deutsch has narrowly conceived evolutionary theory considers as mere vehicles for the replication of our genes our memes and it refuses to address the question of why evolution has tended to create ever greater adaptive complexity or the role that such complexity plays in the wider scheme of things so there is clearly implying that adaptive complexity does have this larger role to play we now see how it is possible for the universe to increase both organization and entropy at the same time the optimistic and pessimistic arrows of time can coexist the universe can display creative unidirectional unidirectional progress even in the face of the second law because this is paul davies says that um so that's really uh kind of a key point uh in this is that um the second law actually is in a sense driving this increasing progress so not only can we have unidirectional progress in the face of the second law it seems that this progress requires the second because we'll see that the second law is basically the natural selection pressure for uh self-organizing systems so in a sense the second law is both uh shiva the destroyer and brahma the creator um a goal of this unifying theory of reality i've been calling the great consilience it was actually a name uh suggested by marco lin who is uh is a fristonian in influenced uh some of this talk um so basically the unifying theory

of reality illuminates the connection between complexity cognition consciousness and cosmic evolution or matter mind and cosmos um and it's really based on complexity science it's a lot broader than complexity science but i guess that name kind of in a way includes everything so it's a unification of major sciences of our time including but not limited to physics biology neuroscience computer science evolutionary theory and statistics and basically it describes complex adaptive systems of all scales in terms of energy and information flows it uses statistical thermodynamics information theory and cybernetics um so we will talk about those a bit uh coming up so if you want to see kind of like a a map of what sort of informs uh this non-reductive theory of everything here you go so i know there's a lot of information so i'll kind of try to simplify this for everyone um there are some unifying paradigms that have emerged so basically this theory is nothing new it's basically evolutionary epistemology um after that came universal darwinism and later uh right now we're seeing this kind of revolution with like bayesian inference and these ideas being applied to everything being informed uh by these paradigms uh combining those paradigms are updating them i should say with uh information from non-equilibrium thermodynamics so we're talking about the thermodynamics of open systems uh information theory so shannon's theory some of you will know that information theory is related very closely to non-equilibrium thermodynamics and basically uh boltzmann's statistical entropy is mathematically the same as shannon entropy or information entropy which is a measure of uncertainty um if i had more time i would go through thermodynamics and the different types of entropy but we don't have time for that but if anyone has any questions i can clarify um cybernetics was basically information theory applied to biology adaptive systems it was the first science of adaptive systems really and uh it's funny while writing this uh i found out because you know i hadn't i had heard the name cybernetics but it wasn't clear how important it was in a talk by uh jim crutchfeld uh santa fe institute basically revealed that uh because cybernetics got associated with like the russian uh like during the cold war kind of like the russian agenda um cybernetics was used to design like heat-seeking missiles uh norbert weiner's theory was almost immediately applied for military uses and because of that it stopped being taught in american colleges yeah it's really important and it was actually emerging at the same time as information theory people were already uh thinking about systems in terms of information flows but it's more than information theory because it's a science of feedback so dynamical systems theory or complex adaptive systems is kind of like an offshoot of that all of these inform what i've called universal basianism not the only person to call it that i had searched that term and found adam saffron's uh paper on

integrated world modeling theory that used that same term but uh john campbell has been a researcher who has done a lot to show this equivalence between evolutionary processes and processes of bayesian inference we're going to talk about that in a little bit um so yeah if anybody has any questions about that but really all of these things origin of life research integrated evolutionary theory um if you haven't checked out the work of these people like everybody here is worth reading their work but uh so from that we get a unifying theory of reality that is based on what i call the evolutionary epistemology universal darwinism universal bayesianism framework that's a long name i'm aware of that but it's just to show that these three theories are basically the same theory universal darwinism says that the universe is evolving at every scale through both competitive evolution and through self-organization which is cooperative evolution among agents that have formed a collective unit so it started with richard dawkins and this concept of the meme and the selfish gene that uh memes uh were these uh cultural units that were equivalent to genes and so we started to see how information was this uh paradigm that could be extended beyond biology and then we had dan dinnett and darwin's dangerous idea really extend this to like everything to self-organizing systems to cultural and technological systems uh evolutionary cosmology came before that it was based on karl popper's philosophy but it was actually invented by cognitive psychologist donald campbell and basically that said that the evolutionary process is a problem-solving procedure that creates knowledge so the information stored in genomes and in uh brains and cultural memory uh that all that knowledge accumulates from an evolutionary process and it's really helpful to think of it as knowledge especially when we start to talk about this in in uh the context of thermodynamics and information because uh basically this knowledge is what allows life adaptive systems to stay far from equilibrium they can resist this tendency towards decay life can uh continue to persist if it can acquire the knowledge that it needs to essentially to find free energy that will allow it to sustain itself um so we'll talk about that in a second more um so universal bayesianism is a bayesian update of the first two paradigms knowledge is encoded so the knowledge that evolutionary epistemology can describe it's actually encoded in biological systems uh in the form of a world model so this model is an internal representation and statistical mapping of the environment which gets updated through adaptation and adaptive learning we're going to talk about that process really the the whole bayesian brain hypothesis and free energy principle is inspired by cybernetics heavily so ross ashby's good regulator theorem and law of requisite variety which we'll talk about was kind of the first talk of systems having models and having to model the environment um so kaufman

says inference is actually quite close to a theory of everything including evolution consciousness and life itself we're not going to go into those bigger implications that would be at the end of the book in the last chapter but basically this isn't just a theory about um well it is evolution but evolution applied way more broadly so quantum mechanics has an interpretation that's consistent with this and bigger problems like the fine-tuning of the constants and parameters that allow for the emergence of life in the universe we can actually uh probably have the best explanation for the fine-tuning problem and the measurement problem of quantum mechanics uh with this theory um we won't talk about it but just so you know next time uh cosmological natural selection by lee smolin is the theory that kind of addresses the fine-tuning problem from a universal darwinism perspective while quantum darwinism uh addresses the measurement problem um so we will save that for next time but feel free to ask questions about any of those so what does this do i have this name poetic meta naturalism i meant to take that out but uh next time we'll get into why uh i chose that name it's actually um it sounds kind of like you know like complex or buzzword or jargony but there's actually very practical reasons for having that name um so this uh unifying theory bridges matter with mind so universal darwinism gives us a picture of the open-ended complexity growth that this whole idea is based on the idea that evolution keeps producing increasingly complex intelligent forms of life excuse me yeah universal bayesianism is uh the link that kind of connects those paradigms with uh consciousness so again we won't get into that today but when uh you understand that knowledge is stored stored in the form of a world model that a system that models itself can experience then we start to get a better understanding of like what conscious consciousness consciousness is it's a it's a mental model so it's not a traditional pan psychic theory even though i know firsthand and levin and some of his colleagues have kind of been flirting with pan psychism this theory would say that's wrong and it's kind of uh reduces consciousness to something trivial um by saying that it's in everything like a proton is conscious because it's a triad of corks that have to be in a certain configuration so there is some minimal level of integrated information um that's not enough uh to get a conscious experience again observer um so saffron's integrated world model theory says that you need a model with spatial temporal and causal coherence i would also argue that it requires self-modeling capacity talk about that a little bit at the end if anyone's interested but this theory is supposed to bridge mind with cosmic so the story starts with the second law um daniel if there's any questions about that like intro let me know yeah i can just proceed could i ask two questions from the chat sure okay the first question is from blue blue wrote do you think that

non-living autonomous agents with multiple competing priorities such as self-driving cars have free will if not what is the defining difference between these systems and humans uh no i don't think they have free will so um adaptive information is a special type of information i do think those things and this is kind of following sarah walker's lead who's at arizona state and she's also sfi faculty now but to have self-driving cars you first need life anything like a self-driving car that's like this information processing system uh if you were to come across a broken down self-driving car on some planet and you saw that you would have to infer that there was also life there because life is part of the trajectory to get to that but i wouldn't say that it has free will because it's just this input output machine uh with a defined set of algorithms adaptive systems are more flexible uh and there okay so in the next next time we'll talk about actually the emergence of agency which occurs with the origin apply for abiogenesis and you basically um because this self-organized system that would be kind of like the protocell is evolving through a mechanism that a darwinian dynamic for self-organization that will explain in a bit it builds up information and at some point there's a phase transition or the information that's getting built up in the system uh basically uh gains what sarah walker and paul davies call informational control so there's a point where information actually starts calling the shots you can see the difference between uh systems that where there is informational control uh so living systems and inanimate systems because for example inanimate systems uh their movement can be predicted with newton's laws so anyone who's taken first year physics uh probably had to do an exercise where you're trying to predict the movement of macroscopic system like a ball that gets kicked or pushed by a gust of wind and what you do is you um you have to draw out all the forces acting on the system and then you can understand where the system is going to go with life you don't need an external force for it to start for example climbing up hill um that's internally generated of course it's not magic the adaptive system the agent has stored energy that it's extracted from the environment so for example humans that have eaten food and there's a metabolic process that's driving this uh still yeah you with living systems that where information has causal power basically the trajectory of the system can't be predicted by these fundamental laws of physics philip ball a journalist that has been talking about this his recent article which caused a lot of controversy between like the reductionists like talking about free will and agency um and there's a lot of confusion around that so people like jerry coyne were like trying to shoot it down and yeah hopefully i can talk about that um i guess we'll talk about that next time because there's a lot of uh confusion about what we're saying when we say something has agency or free will but his

example is if you throw a ball and a bird off a tower you can predict what one is going to do you can't predict what the other is going to do interestingly though i would say that the bird's not completely unpredictable you can predict that it's not going to hit the ground it's not going to splat on the ground it's going to fly away you have to understand the bird in terms of its goals so you have to understand you know the bird and its environment and the information that got built up through evolution and then we can start to predict the bird uh its statistical behavior so inanimate systems you can pretty much precisely predict the movement unless they're chaotic systems which is another story but with life um yeah it's not predictable in that same way with like low-level laws of physics but uh i'm arguing that actually life is more predictable than we thought but you need things like non-equilibrium statistical mechanics for example uh free energy principle um so you can say that you know an organism try to minimize free energy or minimize the difference between uh its model's prediction and reality so it will try to minimize prediction error and that's a way that we can start to describe uh behavior statistically yo is there another one yep can i ask the second question yeah i i know i didn't completely answer that but uh that will require like the all the points that i would get into in the second talk but it's a great question yeah i would have written in a larger margin if i had more time the second question is from joseph clark who wrote uh first complimented your talk and let you know that you were an interesting guy so then the question was how far do you think the universe can organize itself and what does that look like okay great that's a perfect question to lead into the next part and hopefully by the end of this we will have some sort of answer for which is good because i when i started making the beginning of the presentation i was going to get to consciousness and free will and i ran out of time so there's been a lot of talk about things that i won't get to but that we will get to the story starts with the second law um so the second law of thermodynamics has popularly popularly understood says that uh systems naturally become increasingly disordered um that's really just because of the large-scale effects of chance so there are many ways to be disordered and there are relatively fewer ways to be patterned a system that's this collective of particles so this evolving ensemble of particles will naturally tend to move towards a configuration that's completely spread out and mixed up there are no energy gradients so no work can be extracted that's a state of equilibrium so we are going to use this word thermodynamic equilibrium to mean death and disorder so life wants to resist the tendency towards equilibrium it wants to stay ordered uh what's interesting about this is that it applies to all this is kind of like where the free energy principle and for instance bayesian brain hypothesis it frames everything this way so i thought

it was a good way to kind of frame it here is that you start with the second law and any conceivable system any ordered system for it to continue to persist has to resist this uh tendency uh schrodinger the guy everybody knows from quantum mechanics in the in the cat thought experiment he wrote a book called what is life in the 40s and basically kind of solved his paradox like if there's a second law and things become more disordered than what's with life and what's with the biosphere and all this complexity that we see around us and he explained actually boltzmann explained this before him bud boltzmann said a lot to contradict this as well or just didn't follow this to its implications but uh schrodinger explained that open systems uh by feeding off uh free energy in the environment so free energy is just energy that uh you can extract work from so it's ordered energy as opposed to energy that's been dissipated in the form of heat you can't extract energy uh uh useful energy from that energy it's still there it's just spread out and it's it's you can't harness it collecting it would cost more than you'd get out of it so um schrodinger explained that uh if a system can get free energy which he called negative entropy or energy uh neg entropies was called by uh someone later so to understand how ordered systems stay ordered you basically need to understand that they have to constantly be extracting free energy from the environment the moment that they can't extract more free energy is the moment that the system dies and decays to equilibrium so basically uh the second law is not violated by life because in this effort to stay far from equilibrium life it keeps extracting that ordered free energy and keeps dissipating it as heat so and so it's basically exporting entropy into the environment so the order that is maintained is paid for by the dissipated energy which come becomes thermal entropy so there are multiple definitions of entropy so you have thermal entropy which is heat entropy and then statistical entropy which explains thermal entropy but it has a broader domain of application so you can talk about shuffling a deck of cards and you can talk about like the statistical entropy increasing if the deck is ordered and it starts becoming mixed up through the shuffling but the deck of cards the thermal entropy uh isn't changing much at all so there's different types of entropy there's also information entropy which has to do with the um uh the number of messages that could be sent across the channel instead of um the number of states that a system can be in the number of microstates without changing the macro state macro state or the kind of collective properties of the system there's a really neat relationship between statistical entropy and information entropy which i won't talk about here it was really fleshed out by e.t. janes but yeah so information and entropy like go hand in hand um just to give a little more context before we get into the details uh david deutsch and this great ted talk um called after billions

of years of monotony the universe is waking up says if one can speak of a cosmic war it's a war between monotony and novelty between stasis and creativity and in this war our side is not destined to lose if we choose to apply our unique capacity to create explanatory knowledge we could win so that's very different than brian green's story and it kind of gets at the second question that was asked there how how far can this complexity increase go and so this kind of foreshadows that maybe it doesn't have a limit and that seems you know most people think that would go against the second half of their own dynamics we're going to see why that's not necessarily true but you can also see this cosmic war as this fight between order versus disorder life versus entropy and knowledge versus uncertainty or ignorance um but is life's battle against disorder uh good versus evil or yin and yang so i'm gonna kind of argue that these two things kind of have this uh you could describe their relationship as causally dependent and it's kind of like this dialectic between life and uh disorder um and it is through this sort of uh interaction that uh creates progress so we have some eastern ideas mentioned again um but i when i was writing the book a friend practices buddhism called me and told me about this interdependent colorizing idea which i thought was really cool so we're going to come up with this what we call poppers principle which is problems create progress we're going to see that this problem of entropy that this tendency towards decay is what actually forces systems to evolve basically it forces them to search what's called the configuration space for the phase space for the state space to find configurations that are good at extracting energy and that allows it to stay far from so we're going to have some figures soon so this stuff that's kind of abstract becomes a little bit more concrete just out of respect to the earlier request how about you check the time and feel then how fast you'd like to continue however much you want to present and then if you want to take any questions but first just check the time and let just i i i didn't uh i didn't notice when we started so about how long have i been going i would say five zero minutes okay how long can we go we can go another uh 99 minutes okay um i'll try to go through it a little bit faster um but yeah so um but if we use all the time i told you yesterday that i was feeling uh oh i had a sore throat and uh the sore throat's better today i'm hopped up on cold medicine i'm sure i'm gonna i'm gonna pay for tomorrow but um yeah if we use all the time with questions that would be good uh so in our everyday experiment experience things don't naturally become ordered unless someone intervenes rooms get messier buildings erode uh so we already talked about this schrodinger's solution so we can think of life as a game so life has always been playing a game against the second law of thermodynamics there is another narrative that i won't go into but life is also an energy flow

channel so life by existing in computing and trying to stay far from equilibrium is also increasing entropy uh actually the universe of life will increase entropy at a faster rate than a universe without life um which is pretty interesting so it's again not necessarily that life is like battling against entropy in in many ways it increases entropy but there's also this story i guess what it's trying to battle is a tendency towards disorder so it depends on what type of entropy you're talking about so the challenge in the game of life which applies to you as well as other simpler adaptive systems is to resist that tendency so this gives life a goal or a teleology so two algae was associated with kind of this mystical thing but basically we'll we're seeing that teleology is just uh agency it's just a result goal-oriented behavior as a result of the information that's been accumulated over evolution so teleology the book uh tries to naturalize the teleology and just say that they're these goal-oriented systems that behave different uh because they are acquiring information through darwinian evolution so life has to extract free energy that's how it stays in the game of life and to extract free energy it's it's not a simple task so the computational task of extracting free energy uh requires that the uh system acquire information about its environment otherwise it can't find energy uh it also requires that the system model its environment so the information is creating this predictive model this generative model and it is um that model that allows it to continuously find free energy um so we will explain that life strategy is it searches through a configuration space for solutions to the problem of survival through trial and error search uh variation and selection so we we know about evolution there's this genetic variation in natural selection and uh we're going to argue that that process that mechanism is much broader the evolutionary mechanism is actually the mechanism of the scientific method and of adaptive learning um and that mechanism is also a process that accumulates evidence-based knowledge so it's also a process of bayesian inference we're going to explain why so here we have those different mechanisms with different names so it's not included here but uh so if life is a game there are levels to this game uh and the levels we're going to see are these uh evolutionary transitions every time life graduates to this uh more uh complex uh hierarchical system like when single-celled organisms come together to form a multi-cellular organism and multicellular organisms come together to form communities like societies uh that you can view these evolutionary transitions as life graduating to a new level we're going to talk about how adaptive information gets built up through evolution and we're calling that knowledge it's nice to have that word knowledge because this process of an adaptive system acquiring this information is actually reducing its uncertainty about the environment so it's it's reducing its ignorance about all the ways the

surrounding world could potentially surprise it uh so knowledge is a nice word some philosophers like old school philosophers might have issue with that um because in some teachings like knowledge is something that like conscious beings with like awareness have but we're calling any adaptive information knowledge and it's got more of a technical definition because knowledge is that information which reduces uh uncertainty and you could actually quantify this process of evolution terence deakin has talked a lot about how you can do that with uh information measures so just to explain this why is knowledge something that matters like does it have causal power or is it all just this this billiard ball universe where it's like these particles bumping into each other and you can basically use uh fundamental laws of physics to describe like the trajectory of all the particles in the universe this is saying no that uh the information that gets built up through evolution has causal power such that these systems start to behave in ways that are different than what would be predicted from uh something our newton's laws we also have this process of anti-accretion that sarah walker likes to talk about uh but anti so anti-accretion is uh basically we're familiar with uh planets pulling in matter so bodies like asteroids because uh gravity pulls matter in naturally but anti-accretion is when a planet is ejecting matter and uh in many cases what well so the the only time we really see anti-accretion uh is when you have a planet with life on it so right now uh bits of biosphere so humans and space rockets are being sent to other celestial bodies so that's a form of anti-accretion uh sending up satellites you won't see a satellite get sent out if you see that in the universe like you're looking at some other solar system and you see a body uh sent out and put into orbit from a planet that's an indication that life is there that's a biosignature so she says that anti-christian requires comprehenders specifically the existence of physical systems with knowledge of newton's laws what does it all mean um it means we need knowledge uh to uh keep us out of thermodynamic equilibrium how do we acquire knowledge uh science is the most uh is the most salient example of the causal power of knowledge so science uh has uh eradicated lethal diseases built a global communication network called the internet and created weapons of apocalyptic power it's not always good this power of knowledge but um uh we see that there's something about the scientific method which is efficient at uh accumulating knowledge so we're going to talk about why and that's where we get into karl popper and evolutionary epistemology so we know knowledge is important karl popper says we're always faced with practical problems and out of these grow sometimes theoretical problems before we try to solve some of our problems by proposing theories popper stresses that all of science starts with a problem either theoretical or practical um and it is that problem that forces us to

seek out a solution so we can already see why this principle that called poppers principle uh uh says that problems create progress some of these theories that we make to try to solve our problems will be wrong they'll be errors um but the idea with science is that those errors get filtered out by a process of testing your theories and a peer review process which gives criticism so uh knowledge is really uh for knowledge for us to know that some piece of information is knowledge it really needs to be tested and uh we call that evidence-based knowledge or evidence-based information so it's really the only true kind of knowledge because you don't know if something's true until you test it for example um the world doesn't look round from a naive perspective it looks flat so for example um someone who had the first idea you know and i'm sure it was before anybody famous we knew someone had this idea and probably people blew it off but um some things that are true about reality are not immediate immediately apparent from like sensory observation from since data so we really need to test our theories if we are going to be sure about them um so what popper found out was the scientific method is an algorithm he called that algorithm conjecture and refutation and uh basically this is the method of hypothesis testing so you have a problem you believe can be solved you make an informed guess or conjecture to see if your theory can be refuted uh prove false by testing its predictions uh people have pointed out like john uh carl fristen that science is a process of inference so it recognizes statistical patterns and trends in nature that we can use to make increasingly accurate predictions it would not be inaccurate then to say that the functional purpose of science is to generate predictive knowledge and we can characterize that as process of inference which basically means that scientists draw logical conclusions about the way the world works uh based on evidence-based information acquired in the past but with never present awareness that the information can and will lead to to new conclusions and deeper understanding so when we recognize this we also have to recognize that all of our models will have uncertainty so we should never think that our model's right every model is going to be proved wrong in some way but through the scientific process uh the idea is that we get closer and closer to truth so we can never reach perfect truth but in this model you can actually get closer to it so in some senses it's different than like post-modernist philosophy that says there is no objective truth there is an objective truth um uh but we can never know it but we can get closer to it once popper understood that knowledge the knowledge generating mechanism behind science success was conjecture and reputation he realized that human learning which begins at birth and continues to adapt uses the same problem-solving algorithm although the developmental psychology literature called the method trial and error so because

life is constantly presenting us with new challenges such as the need to get from one place to another a problem that inspires babies to learn to walk we must constantly try out new behavioral solutions we can think of these actions as guesses about how to survive or if you prefer experiments in evading equilibrium or predictions for persistence and they will often fail this is dan dinette's idea of life exploring this design space we're going to talk about how it does so through evolution but we're still talking about science here and adaptive learning on abstract level we can imagine a certain practical problem as a challenge one with a solution that exists somewhere out there in the space of possibilities just waiting to be found by someone sufficiently motivated and clever it may take some time but if the possibilities face the space of possible solutions is continuously explored in an efficient way eventually a solution will be discovered so excuse me trial and error learning explores the solution space um just to give you a clear example of how adaptive learning uh is kind of this form of hypothesis testing so whenever we have a problem in life and we don't know the correct solution to a problem in advance um we naturally begin with those potential solutions that are closer to our starting point in the solution space for sampling so for example if a baby reaches for its bottle and barely misses it'll adjust its behavior slightly which minimizes the chances of making an error our instincts may nudge us in the general direction of a solution but the instinct alone is not enough we have to try out what you could call a behavioral conjecture and when it fails uh we try something new so that's kind of like a new theory and it's essentially the old one but with a little twist and if that behavioral solution is remembered then it gets stored for future use ready to be repeated but also adapted if necessary but in humans so the baby example reaching for the bottle let's say it misses the bottle it won't get a reward signal it won't get a little surge of dopamine so if it corrects and it gets closer it gets a little reward signal this is reinforcement learning but basically this adaptive learning process is a process of making theories and eliminating those theories that are errors popper saw that scientific knowledge and the evolutionary process were also connected so not only is science uh adaptive learning uh both of those things are extensions of the evolutionary process so he says in science theories are highly competitive we discuss them critically we test them and we eliminate those theories which we judge to be less good in solving the problems which we wish to solve so only the best theories those which are most fit survive in the struggle this is the way science grows um so that's kind of nice like showing you really how this is an evolutionary process um not only is science an evolutionary process the converse is true evolution is a scientific process so it is clear this is another quote it is clear that this view of

the progress of science is very similar to darwin's view of natural selection by the way of the elimination of the unfit of the errors in the evolution of life the errors the attempts at adaptation which is a trial and error process analogously science works by trial theory making and by the elimination of heirs um i might pause in a second just to see if everybody understands this but i have a few more slides so um evolutionary algorithms uh also use this mechanism but it's called generate and test so i didn't mention this uh the mechanism of evolution is variation in selection so we have these algorithms that are equivalent conjecture and reputation trial and error variation and selection we also have generate and test so in machine learning evolutionary algorithms have this method generating tests possible solutions are generated into an actual solution is found and these solutions accumulate in some memory store while the errors are filtered out and forgotten in science solutions our theories are models that activate predicts the natural phenomenon and the successful ones accumulate in peer-reviewed journals and textbooks we see that these processes are equivalent we're actually going to see that it's this bayesian process uh that adaptation is also a process of uh the model of the organism uh becoming optimized uh decreasing its prediction error so that can more efficiently extract free energy from an itch to stay ordered we get this principle um that all evolutionary processes are learning processes and all learning processes are evolutionary processes conrad lawrence a nobel prize-winning zoologist was also a pioneer of evolutionary epistemology and he made the statement life is a cognitive process if this is true and there's this functional equivalence between the mechanisms driving evolution learning and science then that implies that adaptation uh or the genetic information that corresponds to those physical adaptations are uh equivalent is equivalent to scientific knowledge they're actually the same thing so to kind of make this relationship clear biological adaptation represents knowledge of the environment and the knowledge we acquire through learning and science reflects adaptation to the environment so we see that there's no meaningful distinction between adaptive information and scientific knowledge both allow life to predict an uncertain world control matter constrain chaos and construct order from this order so the very same property that is responsible for organic matter leaving the planet in spaceships so this anti-aggression uh is the same uh property that allows organisms to climb up hill seemingly defying the force of gravity of course uh the laws of physics aren't violated in any way but uh living systems are not uh constrained by the laws of physics the way inanimate systems are so some examples a dolphin streamline design which is a product of the information stored in its genome contains the knowledge of hydrodynamics an eagle's wing design contains the knowledge of

aerodynamics not only can we be certain that engineers see knowledge in these functional structures there should be little doubt that they inspired uh our machines and and without that information i don't think we would have come to those inventions at least not any time soon camouflage on an organism represents knowledge of the environment kind of bridging this idea of evolutionary epistemology that evolution builds up knowledge and that science is an extension of the evolutionary process making a bridge between that and bayesian inference uh requires this uh relationship between adaptation and statistical correlation so physicist carlo revelli which who has gotten really interested in explaining how this information that underlies agency that behavior that we see with living systems gets built up through evolution uh he wrote a great essay it won the foundational questions institute essay contest a few years ago but um this work was based on the work of david wolpert and artem kolczynski artemy kolczynski and uh shows how this adaptation builds up information in the system that information is really statistical correlation between the organism and its environment here's the simplest example it's a bacterium performing chemotaxis so bacteria so chemotaxis is basically a bacterium will uh swim in the direction of food and swim away from toxins so basically it's detecting this chemical gradient that follows so we can see in this example um and this isn't the case for all species but you can imagine that a bacterium that swims to the left when nutrients are on the left uh prospers compared to a bacterium that swims at random with this behavior where it's swimming towards food that is the product of a chance mutation so it's in in some sense it's an accident but when it discovers that solution that organism is more likely to reproduce so that solution gets retained in the population it gets sort of frozen hardwired in so the process of adaptation is a process that builds up a statistical correlation between the system the agent and its environment and so that shared statistical correlation represents mutual information so mutual information is information that's shared between the environment and the organism uh just to explain that a little bit uh this correlation uh basically as this correlation gets built up it's creating predictive john maynard smith cites friedretzky who did kind of the pioneering work applying shannon's information theory to try to quantify evolution so dredskey argues that if some variable a is correlated with the second variable b then we can say that b carries information about a for example if the occurrence of rain is correlated with a particular cloud then the type of cloud tells us whether it would rain so that's from a great essay called the concept of information biology we can see that you have this little rhyming equation here adaptation equals statistical correlation equals mutual information uh and because that uh shared information allows the organism to predict its environment

better this process is also a process of model optimization so the predictive model of the organism as it adapts starts to become more accurate and so this whole process can be seen as a free energy minimization process of course that's information theoretic free energy it's the free energy of for instance principle we're not talking about thermodynamic free energy anyone wants to ask a question or gets confused about that but it's basically uh those things are related and it's something that's not mentioned enough uh minimizing information theoretic free energy which is just minimizing the prediction error of your is what allows the organism to minimize environmental free energy by acquiring information reducing your information theoretic free energy allows the organism to extract energy and reduce thermodynamic free energy it minimizes the free energy in the environment converts it into we kind of see how this is getting into universal bayesianism as it becomes more correlated it gains information about its niche and therefore becomes a better predictor of that niche so that's what adaptation is it's the genome of the species accumulating information that reduces the average organism's uncertainty regarding the so um the genome is a knowledge repository of solutions to environmental challenges that allows an organism with no brain and no mind to anticipate events in the environment so in this theory as i mentioned before consciousness comes later it comes with self-modeling capacity and a bacterium would be kind of a zombie agent it would have agency and it would move with but that is not enough for mind or at least mind defined as consciousness uh if you want to say that mind is just information processing then yes there is mind in these systems but that's very different than saying that the system has a subjective experience and a first person perspective that will come in the next talk but that uh it's it's harder that information stored in memory is integrated into a statistical or predictive model that determines the causal control system's behavioral repertoire or the full range of behaviors an organism is capable of and it is this mapping of sensory inputs to motor outputs that makes the organism move in ways that we normally associate with mind or consciousness um carl fursten in a great eon article says all biological processes can be construed as performing some form of inference i think i already read that but the free energy principle for those aren't familiar we've already explained it but the bayesian brain hypothesis and the free energy principle are the same thing uh the difference is that the free energy principle applies to all other adaptive systems where the other one is just focusing on brains but first and realize that this you know is a model and a process that applies to any system that's trying to resist the tendency towards disorder so any system to stay to continue persisting longer than we would expect uh naturally it must engage in bayesian like learning to deal with environmental

uncertainty how much time do i have left roughly still more than enough okay keep going um i'm definitely more than half through i'd say i'm two thirds of the way through so uh john campbell has this great uh paper called universal darwinism as a process of bayesian inference that i kept coming across as i was trying to understand uh non-equilibrium thermodynamics and information theory um yeah i should say actually i should have gotten the books uh showed you each book but um there's a great book called the origin and nature of life on earth by santa fe institutes eric smith and harold morowitz uh morowitz was a professor of mine eric uh is a friend that uh we would have these foundations of the mind uh guild it was like a consciousness club that would meet at the krasnow institute where i got my phd and um so their book the origin of life on the origin of life in chapter eight actually gets into this information theory and uh this this bayesian inference and describes the origin of life and the evolution of life as a process of bayesian model selection and so that was kind of what connected non-equilibrium statistical mechanics to these to this bayesian brain hypothesis and i realized that you know these theories were kind of converging on the same ideas so john campbell wrote this paper in 2016 he says bayesian inference is an algorithm for the accumulation of evidence-based knowledge this algorithm is now seen to operate over a wide range of evolutionary processes including natural selection the evolution of mental models and cultural evolutionary processes notably including science itself uh when we say that evolution is a process of bayesian model selection and the book i mentioned with eric and morowitz actually characterize evolution as an inference engine uh we're basically saying that when organisms that aren't fit get weeded out that's basically models that aren't predictive getting eliminated so the organisms that persist are the models that were able to predict the environment best so inference and prediction are kind of the same thing uh since the word inference is a little obscure i guess um so just to make this point a little more clear since nature is complex and intrinsically unpredictable the model of an organism is at some point guaranteed to fail when reality surprises it with the unexpected so and that's what the free energy principle is about minimizing surprise is the same thing as minimizing prediction error so an organism gets eaten by a predator a baby fails to reach his bottle a scientific theory fails to explain new data basically when any of these things happens it means that the model used to predict reality is not completely accurate it contains a certain amount of prediction error needs to be corrected so what's the solution try something new the old model worked well enough up to that point so don't ditch it just vary it a bit and see if it performs better if it does not reduce surprise eliminate that variant and try again if prediction error is reduced replace

the old model with the new one so that it becomes the reigning theory and the design template for new variants to be based upon when a model has been updated in this way due to natural selection adaptive learning or experimental testing we can say that knowledge has been acquired and environmental uncertainty reduced so here we have an image um i should say that uh infinity maps uh german software mapping company uh helped me create these images uh oliver wanna i have to give a big thanks to him um this image has a little bit more going on than i would want right now but you could see that um this represents two different mechanisms for learning and adaptation so this one on the top that we're going to talk about first is the normal uh competitive evolution sort of darwinian survival of the fittest while this mechanism over here is the mechanism of self-organization so let's first talk about the normal type of evolution we're familiar with and show that how this process of evolution that works through variation and selection actually uh generates predictive information and creates a model which is which encodes knowledge of the biologically relevant regularities in the environment so first we have this this is a bacterium over here and so this is the simplest model um you have an organism that's an embodied theory or a best guess about how to stay far from evolutionary epistemology sees organisms themselves as embodied theories so evolution is testing these different models uh it basically reproduces and we know that because there are going to be um heirs in uh in that process uh the genome will get changed slightly and you will get a generation of progeny that all are slightly they have slightly different configurations some of these configurations will be better at extracting energy in predicting the environment than others and the ones that aren't good at predicting the environment they get filtered out by natural selection so here we see design four is the winning design because it's the best predictor and it's the best predictor because it's the best extractor of energy so that uh design that that organism gets to reproduce and then you have another generation uh of designs that are slightly and this process is an iterative process just like science and hypothesis testing so um over generations you will start to get uh designs that are better adapted to the environment which we've explained means they will be more correlated with the and being more correlated means being a better predictor so the knowledge that's left over after this process is the adaptive information stored in genetic material that reduces environmental uncertainty and you see that this process ultimately lead to the evolution towards the the best possible design so it's it's an optimization a lot of people don't think that evolution is optimizing anything there's been this you know kind of philosophy that evolution does just good enough but we actually see that in terms of the computation that it does to stay far from

equilibrium that it's actually optimized for being almost maximally efficient with its use of energy it approaches something close to what's called the land hour limit which david wahlberg at sfi is written about but uh life is much more efficient uh with computation in terms of energy uh supply needed compared to all of our technology so the human brain can do uh solve problems that even our best like ai systems can't solve and it does so running on uh the equivalent of one household light bulb so we'll come back to this self-organization oh well actually we'll get into that right now so this is the normal uh process that we think about when we think about evolution generating knowledge predictive knowledge and so that's called phylogenetic learning uh phylogenetic refers to these like generations so learning is generational the organisms don't really learn because they don't have brains they're not able to update their model in real time so learning is at the level of the population and for learning to occur organisms have to die before brains basically you have to have this uh competition uh for there to be a learning process however there's also self-organization when these organisms don't compete but actually find a configuration find it discover collective configuration through their somewhat random interactions with each other and when they find this collective configuration through trial and error basically the configurations that are better at extracting energy will be the ones that are retained donald campbell father of evolutionary epistemology uh explain this process this by which self-organization works as blind variation and selective retention or we can think of it as selection for persistence or stability so fitness kind of there's a dawkins quote i think i come out where richard dawkins said darwin's survival of the fittest is really just a more specific instance of a more general law of survival of the stable so when we think about being fit at the beginning we're talking about just being uh stable being able to find energy if you can't do that it doesn't matter whether you have agents that you're competing with or not because if you can't get energy it doesn't matter if there are other people competing for it that you won't survive you don't need replication with variation to evolve that's what this is this model is saying um and stuart kaufman from the santa fe institute has been talking about the importance of this marriage between darwinian evolution and but basically uh because there's not this replication process this is how evolution occurs so you have a system that's a collection of components parts these parts can be molecules or they could be agents and when these molecules are pushed far from equilibrium by a flow of energy of course humans are eating food and metabolizing energy so they're being powered by energy flows as well the system blindly explores various configurations in the space of possible designs via trial and error so this system will move through a series of configurations

and those configurations that allow the system to extract sufficient energy to stay out of equilibrium will get retained while the configurations that don't allow for energy extraction will get filtered out either the system will collapse and have to try a new configuration or it'll just revert to the the previous stable configuration and it will continue trying to explore that configuration space um i should have said like here's a good example like here's a depiction of this configuration space we're talking about life exploring the design space so basically you have a a designer configuration and then you get these uh different this exploration of different configurations that are very close to the configuration you started out with you know i was thinking about giving a talk on the origin of life but i couldn't get into all this cool daisy and stuff that's what the first part of the book is about what's really interesting is that we're finding that as mentioned is a darwinian process so there's actually a name for this dissipative adaptation and it shows that this evolution towards these more stable configurations dissipate more energy when this happens at the level of organisms that we're talking about it shouldn't be surprising that it dissipates more energy because as the organism evolves to be better at extracting energy all that energy it's extracting to maintain far from equilibrium is being converted into thermal entropy all self-organizing processes are really dissipative adaptation processes at different levels so this process philosophers would say is multiply realizable so this process can happen at different scales and that's exactly what we see like right here the same process so you would have a society like exploring different governance systems and it doesn't know which government system is going to work the best beforehand it tries it if it doesn't work well like the fall of rome um it will collapse and then it will reassemble just to wrap up the origin of life so we have this process of dissipative adaptation and if it's being pushed by the flow of and it's doing this process here where it's finding increasingly uh stable collective configurations then um uh basically what will emerge is an auto catalytic set we'll define that later but basically uh it's um i'm using a lot of terms for philosophy i was gonna say like it's the beginning of an autopoietic agent so like a self-maintaining agent in its simplest form is a chemical system that produces outputs which are then fed back into the system so that the system can uh self-amplify it's basically creating all of the molecules uh that it needs to grow it's thought that this process you know occurred the auto catalytic set is kind of the precursor for the first cell so it's kind of like a protocell and um it was thought to occur uh around hydrothermal vents which are under water volcanoes there's all this energy flowing up from the the hot core running through geochemical energy and there's these rocky pores in the rocks around hydrothermal vents provide something like a a membrane like a lipid membrane

to hold the components in a mark off blankets so that the pores and the rocks were the mark off blanket for the auto catalytic set and eric smith and marwitz showed that feeding off inorganic inputs like carbon molecules like not not anything not organic chemistry where it's like life but just like simple like carbon hydrogen and oxygen feeding off those inputs um basically the metabolic process uh it's called the reverse krebs cycle so uh that process allowed uh the first autocatalytic set to self-amplify through this through this process so if this process is allowed to continue um further self-organization may eventually lead to the emergence of an organic computer that can process integrate and store information about the world so what dissipative adaptation is doing is uh as it's doing this blind variation and selective retention process and it's finding more stable configurations those more stable configurations are also predictive of the environment so this dissipative adaptation process is this auto catalytic set modeling the surrounding energy landscape so i'm arguing that this is where modeling begins it even begins before what we call life like rep self-replicating life and it's dissipative adaptation which is a of modeling the the energy in the the energy sources uh this isn't just one continuous process there are these leaps that are phase transitions basically we've heard of phase transitions and physics and chemistry uh courses where you have like water being heated up and then transitioning into a gas or the reverse when a gas becomes frozen uh you get a state of order that emerges these uh transitions towards uh these configurations that are good at extracting energy those uh have also been characterized as phase transitions so eric smith who i mentioned before herald morwitz they have this phase transition theory of life which basically explains how um information gains control in these systems and sarah walker and paul davies work on this basically show that through this of blind variation and selective retention information gets built up in this auto catalytic set and at some point there's what they call algorithmic takeover uh or informational control and the system becomes an agent that can start steer and control itself in ways that wouldn't be predicted by physics so a living system is very much like we didn't talk about dissipative structures but like tornadoes whirlpools those things emerge to dissipate an energy living things uh because of the process of metabolism we are this cycling system this thermodynamic energetic loop uh but uh what's it gonna say about that um so yeah we are like the storm but we do have control over this storm we're a storm with control so it doesn't really make sense going back to sam harris's quote that you are the storm you do not control it it's literally the information that gets built up through evolution that allows you know us being this dissipative structure like a storm to start controlling its own behavior um so the last part is just an argument for that second question

why should complexity increase we know that this evolutionary process where you build up this model uh so we would think that this after these generations of evolution this organism is going to be well adapted to its environment but we know that adaptation doesn't mean becoming more complex always so sharks and crocodiles are cited as species that haven't evolved much at all in many millions of years cave fish actually evolved to become more simple so fish with eyes got isolated and like an underground environment where there was uh need for sunlight and uh so fish like processing light like eyes didn't have any survival advantage so basically fish lost their eyes uh over many generations of evolution becoming simpler so if that's true why should evolution increase complexity the answer to that it comes from this understanding that evolutionary epistemology says that a species adapting to a niche is like a scientific theory being tested for viability species will evolve not to become increasingly complex but to match the complexity of the environment a sort of simplest solution to the thermodynamic problem of staying far from equilibrium so einstein famous we said a theory should be as simple as possible but no simpler the same could be said about an organism you want it to be as simple as possible while it accomplishes its task of extracting energy it has to be complex enough to be able to extract energy in a competitive environment a bacterium doesn't have to do it doesn't need intelligence any extra information processing higher level cognition would be energy wasted that could be energy that allows it to stay far from equilibrium for longer and replicate more so it's not the case that everything will evolve towards higher complexity the reason that we do get increasingly complex species over evolutionary time in a way that i'm saying is inevitable so there is a progressive evolutionary process that leads to higher forms it has to do with uh niche emergence so so basically a species will evolve to become as complex as its niche as its environmental niche but there will be increasingly complex niches that emerge for reasons that will be explained but i will pause there before i get to that which will be what we end on there are no other questions right now so if anyone does have questions how about they can feel free to ask it during this last section give you a second to catch your breath and i know it's been a ton of information like my goal is that you know since this is being recorded it'll be something that's up um hopefully i've cited enough literature that people can explore this stuff themselves and see that like okay it's not this guy that's like uh claiming to have like a theory of everything this isn't my theory it's a synthesis that's emerging and um uh really the story has been there for quite some time and the cybernetics pioneers actually a lot of them were you could call them like cosmic teleologists a lot of them did see this continual increase in complexity of course there's nothing mystical

about it or supernatural about it it's a mechanistic process that doesn't change the fact that nature seems to have these built-in goals that emerge from the fundamental laws and constants the parameters the the fine-tuning but it's a completely natural progressive teleological evolutionary and so i'll just and you know this is finishing up it's along the same line so that's good the questions can come after this and i know it's been quite a lot so i will just try to sum this up simply um two important principles from cybernetics these came from ross ashby who also created the principle of self-organization which was really the first time people took self-organization seriously so we think of complexity science as being the science of self-organization but it really goes back to cybernetics um i think the term i think kant might have talked about it but it was ross ashby who tried to you know give it a make it a principle and a formalized concept so there are these two concepts that really i think anybody interested in like coffers and spacing brain hypothesis free energy principle should be aware of because all of the way he talks about like evolution comes from this ashby's work the good regulator theorem it was also updated there there was the internal model principle which came like a decade later good regulator theorem came these both these laws i think came in the 50s in the 40s or 50s and then i think in the 70s the internal model principle was created but it basically says that any system that regulates or controls another system must have a model of that system otherwise it can't regulate it so it seems pretty obvious but this talk of models kind of starts here that is very familiar to like anyone interested in like the free energy principle this can be applied to inanimate systems i mean controllers are like house like thermostats and stuff but ashby was really interested in adaptive systems so he was thinking about evolution so applied to evolution an organism must model its environment uh if it's going to extract free energy and an organism is a model of its niche like a key is a model of the lock it opens so when we talk about these models it can be kind of extra kind of abstract i think this a key is a model of the lock it opens really explains how an organism fits into a niche you could also say like a hand fits into a glove but it really the organism is a model of its environment so natural selection is essentially an information channel that through evolution is pumping in from the external world from the environment into the organism such that is encoding the structure of reality and encoding the structure of the world and its design uh include uh uh encodes information about the outside world so it's kind of a new way to think about evolution as this process of the universe is modeling itself and waking up through this process it's pretty interesting um kind of psychedelic so um the law of requisite variety is just telling you about the sophistication of that model so how complex does the model have to be for the organism to

stay far from um ross ashby's law says so john naughton researcher at cambridge uh kind of summed it up this way in colloquial terms ashby's laws come to be understood as a simple proposition if the system is to be able to deal successfully with the diversity of challenges that its environment produces then it needs to have a repertoire of responses which is at least as nuanced as the problems thrown up by the environment so a viable system is one that can handle the variability of its environment or as ashby put it only variety can absorb variety so it's what we said before with an organism evolves to become as complex as its environmental niche requires it has to specifically have enough behavioral responses which map onto internal states so it has to have enough cognitive states respond to the number of challenges the environment presents challenges in its ability to extract energy when you get organisms and you get other uh you get other agents other organisms then what the organism has to model is much more complicated because it doesn't have to just model the energy source in its environment like to get food it also has to model other modelers other cognitive agents so yeah basically any adaptive system has a behavioral repertoire and a repertoire of accessible mental states or for things without brains computational states and the number of states that the organism can access uh should match the number of challenges the number of states that could surprise the organism by the um so a cat must have at least as many states as the ways the mouse can invade it and the mouse has to have enough behavioral states to get away so they model each other a swordsman like a fencer must have many blocks as his opponent has attacks and as we said because bigger responses map onto unique internal states the loss of an organism must have as many accessible states as required by the complexity of niche so we already said this uh organism is some simplest solution to the thermodynamic problem of seeing far from so a well-adapted species represents a biological solution to an existential dilemma it's a sort of living evolving scientific theory about how to most efficiently extract the free energy the lifeblood of existence out of a particular niche some niches present a changing variety of challenges that must be adapted to you while others present hardly any at all so almost done here i know it's a lot um hopefully this can kind of wrap up the ideas and kind of show how they're all connected um but this is still on this question of how complex is the universe uh how complex can it get so olivia olivia judson wrote this uh published this paper called the energy expansions of evolution it's really nice people should check it out uh so she says the history of the earth life earth system can be divided into five energetic epochs each featuring the evolution of life forms that can exploit a new source of energy these sources are geochemical energy sunlight oxygen fire and the first two are present at the start but

oxygen flushing fire all consequences of evolutionary events and then she makes this statement by the way red is quotes if people haven't uh figured that out so these are her words i forgot to put quote marks um so no category of energy source has disappeared this has over time resulted in an expanding realm of the sources of energy available to living organisms and a commitment uh increase in the diversity and complexity of ecosystems here's the way to think about it um so taking the thermodynamic perspective and looking at organisms as energy flow channels uh we can think of each niche on earth as a sort of energy slot for a given species and an evolving population of organisms efficiently searching the solution space will eventually just by chance discover a solution to a thermodynamic problem that it didn't know existed the discovery of a novel energy source or energy extraction technique is both how a new niche and a new species come into existence so phylogenetic learning which we saw in chart will naturally lead to speciation because organisms will stumble upon new ways to exploit thermodynamic niches that were previously inaccessible to life purely for design reasons we're going to give examples on the next slide so as the biosphere accumulates knowledge through phylogenetic learning not only is the solution space corresponding to one kind of thermodynamic problem being explored there's a growing problem space each with its own unique solution space so in an evolution so there's an evolutionary trajectory that's not determined in the strict sense envisioned by laplace where like there's no freedom in the future it's like set in stone but there is a sort of uh statistical determinism what i've been calling a non-equilibrium statistical determinism that guarantees that basically the inevitability of this emergence of a distribution of far from equilibrium attractors of varying varying degrees of complexity so each species can be seen as a an attractor that's exploiting this niche and you get this uh basically this diversity where you have organisms uh exploiting like the the niches that were kind of the easy ones to exploit on earth like the geochemical free energy we mentioned with the origin of life and then sunlight was a little bit of a more difficult challenge to extract sunlight it took some time for evolution to be able to do that so uh extracting energy from the hydrothermal vents uh source of energy is really simple because the energy is already like flowing through the system it doesn't have to do much work to extract sunlight you have you know a moving sun it's a little bit more complicated but not nearly as complicated as organisms that eat other organisms so heterotrophic organisms because they will have to model not just like a sun moving predictably through the air but have to model all these other agents many of which are trying to eat them to kill you um so basically so answering this question why increasingly intelligent species emerge we started off with reductive autotrophs

those organisms around hydrothermal vents they were a solution to the problem of how to extract free energy from geochemical gradients um photosynthetic bacteria the ancestors of plants for the solution to extract of how to extract all the solar energy that was flowing through the planetary system so that was a big discovery made by life is that you can get energy directly from sunlight heterotrophic organisms organisms that eat other organisms where the solution to the problem of how to extract energy from life itself once life starts having to model life other agents with causal power and adaptive behavior the computational task of extracting free energy gets increasingly difficult as increasingly complex species arise so by virtue of having to model each other the complexity of some species gets ratcheted up by what's known as an evolutionary arms race the lion is evolving you know lions are producing offspring some of those lions will be better at predicting the gazelles movements the gazelle will also be creating offspring and some of those better at predicting the lions attack the ones that are better at predicting are the ones that get to uh stick around so you see how you have this ratcheting mechanism that ratchets that ratchet is up complexity um there's actually a principle called the red queen principle uh that says that for an organism to simply stay in the game of existence certain organisms a certain species they will have to become increasingly complex and that comes from uh alex in wonderland basically there's a scene with the red queen and alice's they're they're all running as fast as they can and alice is like you know i'm running fast as i i'm i keep running but i'm i'm going nowhere like so they're they're all running but they're they're not making any distance they're staying in place and the queen says something like oh you see here for that to go somewhere you'll have to run much faster than that the idea is that for a species that has to deal with this complex environment just to stay where it is just to stay in the game of existence it has to become increasingly complex it has to keep adapting one thing that i wanted to get into more but i'm not going to is this idea of empowerment it comes from the computational neuroscience literature it's similar it comes from daniel palani but it's very similar to harvard's uh alex wisner visner gross he has an idea called causal and tropic forcing but basically these people are arguing that well visitor growth is arguing that a causal entropic force uh means that like basically with like evolution you'll get a system systems are trying to maximize the number of states they can respond to so the maximize the number of states they can respond to is mirrored by their internal diversity so systems naturally try to increase the number of states and it's this ability that allows for intelligence because if you have more states going back to this law of requisite variety you can respond to more challenges not every species is increasing empowerment but you will get the

emergence of increasing uh increasingly complex species because uh each species serves as food for a potential new species um you will get increasingly complex species that emerge over time and they will be increasingly empowered meaning that they can respond to more environmental states and that they have more accessible mental or cognitive states um i explained some of this in the last slide but i'll just explain a little more so open-ended niche emergence why is the evolutionary record shown a trend of the emergence of increasingly complex forms because when all conceivable niches on earth were filled that wasn't the end new species created new niches because the free energy spot they provide is themselves their food cybernetics guy he's kind of like the last living cyberneticist kind of um i'm sure there are other people that call themselves that but he's kind of kept that title even when it like wasn't trendy um he says it is well documented by evolutionary biologists that ecosystems tend to become more complex the number of different species increases and the number of dependencies and other linkages linkages between species increases he cites eo wilson who recently passed away who did a lot of work that that kind of inspired a lot of the concepts in the book um not just this niche emergence but also talking about super organisms so collectives like ant colonies self-organizing systems that all scales but he said not only do ecosystems contain typically lots of niches that will eventually be filled by new species but there is a self-reinforcing tendency to create new niches what does that mean it means uh the biosphere certainly ecosystems are themselves auto catalytic sets seth lloyd he's talking about these chemical auto catalytic sets but it's i should have had this on the earlier slide but it'll be a good way to kind of show the equivalence between ecosystems cethloid says auto catalytic sets of reactions are powerful systems in addition to computing they can produce a wide variety of chemical outputs in effect an auto catalytic set of reactions is like a tiny computer controlled factory for producing chemicals some of these chemicals are the constituents of life now stuart kaufman big you know one of the main names who's kind of this whole talk he's kind of put these ideas out there uh since like the early 90s well actually going back farther than that but he started writing books in the 90s and really gaining exposure for the santa fe institute he's written a recent paper well semi-recent niche emergence as an auto catalytic process so ecosystems are auto catalytic auto catalytically closed self-sustaining reaction networks so they're auto poetic agents that reliably drive up biological diversity and complexity as they self amplify and evolve so when you think about it really all integrated networks of adaptive systems functions out of auto catalytic sets there's been a lot of papers showing this economies in a sense are autocad sets including the social organisms we call

societies or civilizations so these networks are common because they emerge spontaneously when many organisms repeatedly interact with one another and discover synergistic as nature often pressures them to do so we're at the end um almost uh we've explained why complexity evolution will not increase the complexity of every species they will become adapted to their niche they will match the complexity of the niche however because new niches are always emerging because species act as food for a potential new species because of that process you will get this increase uh you will get species that emerge that have a higher number of uh a larger repertoire of mental and behavioral states so the law of requisite variety really explains why more complex environments create more complex niches and this is open-ended there's really no limit to how far this process can go especially since humans who are the pinnacle of this process we're not you know a lot of people have been against these kinds of views because it seems like it's saying that humans are superior we're superior maybe computationally but not superior in any other sense we're actually part of the biosphere which is this integrated interconnected system it's really you know it doesn't make sense to think of us as like superior to nature because the whole biosphere is an organism that we're just sort of an organ for uh we're basically the nervous system of the planet the cybernetic uh global system that people have called gaia some people don't like that name i think it's a perfectly good name that's why uh complexity gets ratcheted up but the final uh i guess uh part of this complexity increase story is that not only do you get increasingly complex species uh those species will interact naturally to form higher level adaptive systems which are collectives i call this the principle of recursive self-organization based on ashby's principle of self-organization what i'm arguing is that when you have a biosphere with these agents and you have some are simple and some are complex naturally those agents will interact when they interact they will in many cases discover synergistic collective configurations so configurations that make it easier for each agent to extract the energy they need to stay far from equilibrium why do interacting agents link up to form stable holes for the exact same reason that molecules with the right chemical diversity will form stable auto catalytic reaction sets when pushed by a flow of energy working collectively allows the whole system to extract more free energy with less work we can that's the whole point of synergy is that working together makes your workload easier because there's a distribution of labor and it's it's really nice um that you know this distribution of species with some simple and some complex uh that's really what you need to create this like uh cybernetic adaptive system at the level of a planet because a system uh needs diversity among its components in structure and function for there to be this distribution of labor labor

which makes it this collective whole this synergistic uh holistic unit um it's actually good that evolution in adaptation isn't increasing this the complexity of each species like each species isn't becoming intelligent like roaches aren't becoming like more and more intelligent over time with no limit um the biosphere wouldn't be stable if that were the case uh it's really uh ecosystems need this diversity among components just as a car with made of all engines isn't going to be functional or our organism made of all brains you need a variety of parts some simple some complex because they all do different functions um so nature promotes cooperation collaboration and synergy because it is thermodynamically beneficial for all parties synergistic collective configurations will eventually be discovered by many component systems that are exploring various states or configurations through the blind variation and selective retention mechanism mentioned organisms only compete until they finally figure out what that working together makes everyone's task easier and that goes for humans too so it's kind of a moral lesson for everybody the nations have to work together we shouldn't um get rid of the nations there shouldn't just be this evolution into this like global government because the nations are actually uh provide this diversity that we're saying is important uh because you need this uh diversity of component parts to create a division of labor there's strength in diversity this is different than the evolutionary picture of survival of the fittest whereas there's like this idea that the stronger the most intelligent survive this is saying that that's not true it's the most adaptive that survive has nothing to do with strength or intelligence sloths have been around for many millions of years despite being sloths being slow because they exploit a source of free energy that's not being exploited by another uh species so going back to this kind of global message you want nations to retain their identities but you also want them at the same time to come together and to work synergistically so you basically want something like a global system but there has to be the optimal balance between centralization and decentralization neither approach will be good on its own there needs to be this balance as long as the biosphere is creating a growing variety of complex adaptive systems some of those will interact to produce higher level complex adaptive systems which will come together to form even larger ones this process continues uh at higher scales so i didn't go into this uh left out a kind of a neat picture that shows a brain and then the planet and it's like information networks but uh i do i i hope that you know talk about gaia and global brain will become uh will be taken as seriously as it should be uh they really only weren't taken seriously because there were mystical notions associated with this word gaia which wasn't even james uh lovelock's uh fault he act the name guy came from his neighbor who is actually the author of

lord of the flies william golding so blame him for the for the flowery metaphor with gaia um but yeah the the biosphere is a cybernetic organism and we are forming something like the global brain and a self-replicating biosphere is essentially the next step and when we are trying to terraform planets like mars that can be seen as the biosphere replicating for humans to at this point in our stage of development we would have to kind of convert mars to something like a biosphere to have like oxygen and so you can see how it's replication with variation you get the same sort of thing but because it's a different planet there's going to be a variation it's not going to be exactly the same but that's how the evolutionary process continues i should have showed this earlier this is just showing how like eukaryotic single-celled organisms like amoebas will come together to form communities like a slime mold these multicellular organisms come together to form colonies you see ant colony ants actually really do form a super organism you can see here they actually build a bridge out of their bodies so that some ants can like climb across that bridge so it really makes sense to call it a super organism or at least like a super adaptive system or like a meta meta system um because without it we can't make sense of this collective behavior that's functional um of course human civilization we again see this hierarchy now it makes more sense uh we see that these evolutionary transitions oh yeah these are called evolutionary transitions sometimes i use the name meta-system transition because if we were talking about the brain i would also talk about like uh the emergence of a prefrontal cortex which is like a high level controller um that's also like a meta system transition so meta system transitions are a little extendous where it's not talking necessarily about evolutionary transitions but these other transitions start to model themselves um so here we see uh how evolution has created these different memory systems these knowledge repositories so it's that process of recursive self-organization creates new memory systems when you get go from single cell life to multi-cellular life to life with brains uh it's uh these are in revolutions in information storage machinery so you can see how the information in the biosphere gets accumulated through an evolutionary process are the final points yeah so knowledge is powers not just the hollow buzz phrase of the digital age it's true in the most fundamental way uncertainty reducing information is life's first and last weapon in ongoing war with disorder without knowledge life cannot exist for more than a moment much less colonize the galaxy and beyond this suggests that setting out on the path toward cosmic superiority is not a choice that intelligent agents like homo sapiens make upon careful reflection nor is it just some quirky ambition we stumbled onto by chance this is the main point humanity's collective desire to transcend mortality and expand outward into space so

apparent from our current scientific and technological endeavors spacex uh this doesn't emerge as an accident this is inevitable consequence of the fact that continual knowledge acquisition is a fundamental biological imperative if life wants to persist and stay far from equilibrium at some point it has to get off the planet because the sun going to die and uh it's these sorts of problems these existential problems which uh create progress because it's our awareness of this problem which uh forces us to uh search the configuration space for the solutions that solve the problem so as natural selection pumps information from the inanimate world into life nature begins modeling itself encoding its own structure and the universe begins to wake up where the cosmos come alive not metaphorically but literally this is ray kurzweil's uh image from his book the singularity is near you can see that he sees this he calls this the the the destiny of of life in the universe and he's not shy about using that word um again it's not determined in a strict sense of replace this uh kind of a statistical determinism that creates these uh series of attractors you've seen famous people uh so you know some people see that idea and they'll have been reading brian green and listening to all these reductionists and think that's not a scientific view well uh that's not true and um i hope to see more of these people speaking out about it so christoph coke in confessions of a romantic reductionist says the rise of sentient life within times wide circuit was inevitable teohara deschard in who really came up with this don't have time to talk about him french philosopher and jesuit you should look into is correct in his view that the islands within the universe if not the whole cosmos are evolving toward ever greater complexity and self-knowledge to be clear he's not saying that earth had to bear life that you know primates had to walk the african glass lands but he says i do believe that the laws of physics overwhelmingly favored the emergence of consciousness the universe is a work in progress such a belief evokes jeremiah i don't know the same word for many biologists and philosophers but the evidence from cosmology biology and history is compelling so there's some uh advanced blurbs for this book uh if you want to pre-order it um it comes out in june uh but every uh order that i get from now until then will count towards my first week sales total so that will help me get on bestsellers list if you want to program it now and you email me um i will send you a copy soon of this novel i also wrote called road to omega i was pretty busy in the last five years yeah i went into a cave for like four or five years but this uh it's a vehicle for the science and phosphate about kind of like an anti-iron rand at this shrug maybe the opposite phosphate um and the road to omega sub stack kind of turns his philosophy into kind of an effort to save the world with uh science epistemology and blockchain technology uh blockchains are part of this self-organization process and that's the

longest talk i've ever given that's the longest i ever talked at one time probably my wife will probably say that's not true but i'm tired so let's open it up to questions and thank you for your immense patience it's awesome to hear and think about it so there are two questions in the chat first is from dean dean asks how would we explain caring through a lens of free will or no free will would free will deniers say there is no caring when it appears that agents can or do fully care yeah so great question unfortunately i don't have much to say without saying the things that i was going to bring up in the next talk really explaining what free will is i guess i should just say right now uh kind of a brief explanation of what i mean by free will so i've i hope i've made the argument that agency emerges uh in adaptive systems that start off as biological organisms and um to the person itself driving car thing i uh uh just to mention that real quick i don't think they're agents but i do think uh it's possible to create systems with agency um for reasons that i haven't gone into uh it seems important that the systems have a certain type of architecture so not like a standard computer as a von neumann architecture and it's not uh distributed or integrated in the way the brain is so it's not integrating much information so these systems won't have causal power they won't be agents um neuromorphic hardware has been a lot of people think that's going to allow machines that can think and perhaps even be conscious um i think maybe that's the way to go uh it's probably a lot more complicated than that it's a step in the right direction so agency is real it's uh due to information gaining causal power in a system and starts it starts steering the system so you'll see like a bacterium performing chemotaxis but does it make sense to say that the bacterium has free will when it's going this way in that just because it's not predictable in the way an inanimate system is so i would say that they have agency but they don't have free will what uh free will is and it's not my idea other people have kind of been talking in this way kevin mitchell uh has a book neurogeneticist uh he's coming out with a book called agents which is all about agency and free will but um the idea is that we are agents with causal power but um we're sort of uh um our responses like are basically uh programmed responses uh that are responses that occur automatically uh that are informed by this uh information that's been built up through evolution and adaptive learning so a lot of the time we go through the day and we're not really thinking about things like driving or like waking up and like you know going to the fridge we're not putting conscious thought into that so that wouldn't seem to be to me you're you're kind of acting on autopilot so what free will is is a higher level of control that emerges with something like a prefrontal cortex in a global workspace basically free will allow a prefrontal cortex allows an agent to override its uh instinctive and automatic behaviors and so free will is

something that uh that we have but only if we're exercising this higher level of control it's associated with cognitive control or executive control effort full control and basically if you have a healthy functioning prefrontal cortex you're going to basically the conscious mind is monitoring uh consciousness is a monitor uh so it's it's monitoring uh your uh behaviors and if it uh detects that there's some optimal behavior for whatever reason like it's just automatic kind of instinctual let's say someone says something that makes us angry and we respond immediately with like pushing them or something uh that's not a wise uh move and that maybe may happen when we're fearful because of the amygdala response and we just have this automatic behavior um but uh free will would be when uh the conscious mind overrides that process the bet actually the free will studies that everybody cites as saying that we don't have free will he actually never said that he said free will is in this veto power like maybe some of our voluntary movements occur without conscious thought but it's our ability to override those movements uh that is the source of free but there was more problems with that actually um the studies have coked at a study showing that uh basically that free will um readiness potential which is like a spike in electrical activity that occurs before you do this voluntary movement this movement you think is voluntary and everyone thought that this meant that you're not really making a conscious decision that you could see this spike in brain activity earlier than the person thought they made the voluntary voluntary decision but a new study has shown that that's only the case when we're making like arbitrary decisions and when they had people making a decision about when when to donate like a large amount of money to a charity you actually saw this readiness potential disappear and so it seems like deliberate thinking which is still this like prefrontal cortex and there's this distinction between access consciousness and phenomenal consciousness that i didn't go into that i would in the next talk that adam saffron and integrated world modeling theory really gets at but the idea is that the prefrontal cortex does allow us to override automatic behaviors and that is a sort of freedom that we have and we also lose it uh in cases of schizophrenia you see impaired uh prefrontal cortex activity and drug addicts you see impaired activity in drug addicts you see they do repetitive behavior even when they know that it's not beneficial to them they will still engage in that so they're kind of stuck in this loop for this unhealthy attractor with schizophrenia there's patients that report being pushed around by forces that are beyond their control i think that's because when the prefrontal cortex is uh basically deactivated um people lose this sense this ability to override their automatic behaviors and i think that's what those people are experiencing this sort of loss of this higher level agency that we've called free will um another interesting just the final

interesting thing to mention about that is uh cotard's syndrome cotard's delusion is this delusion where people think they're dead they think they're um like ghost or just not alive it's really weird a lot of people die from starvation because they think they don't have to eat because they think they're already dead um you actually see from narrow imaging studies they've showed uh like impaired activity in the prefrontal cortex so maybe these people who think they're dead have lost this high level agency and and they feel like they're ghosts or something um as far as caring so um caring is a part of like emerge through evolution because you do have this benefit to cooperating with each other this self-organization process that we talked about that happens naturally because basically you find that when you work together you minimize conflict and align it you align interest and so caring altruism empathy all of those things are as natural uh and as darwinian as anything else but you have to add on to evolutionary theory all of this stuff about self-organization and cooperative evolution as far as you see he said something about why people uh who uh don't believe in free will i forget the exact point about the caring what was that um when sorry what was that what was the first part of it um how would you explain caring through a lens a free will or no free will would yeah no it's it it flies in the face of like uh our our everyday experience so yeah um i the the amount of contradictions and like logic that doesn't make sense when you go down that path of there being no free will or agency you just you just get into absurd territory it's i think when we look back it's gonna look quaint that for so long that we thought that everything was determined in this strict and it's really a big issue because these people who believe that if they believe it if they take it to heart they're experiencing cognitive dissonance at every moment you know and the people who argue against agency they talk about religious people having to compartmentalize to deal with you know the the real world but anybody who believes that we don't have agency has to compartmentalize in the same way uh funny little conversation between sam harris and his uh wife um uh annika harris uh who wrote a book called conscious that is a great accessible book but it leaves out like everything that i've discussed and that i would discuss about like the bayesian brain hypothesis uh global workspace theory doesn't really go into integrated information theory in any depth so like all of the things that they're missing that do explain a lot of this stuff uh can be found in like modern neural science so we are making a lot of progress towards you know understanding the hard problem of consciousness so subjective experience but also a free will and but it's funny because they're having this conversation in the podcast and they're discussing whether since a belief and no free will as mentioned that new york times article i think 2011 showed that they have less

moral behavior and it can you know cause depression they're discussing whether they should tell their daughter about free will or not that she doesn't have any free will the discussion is like should we tell our daughter that she doesn't have free will but the actual discussion implies that they believe that they have some choice whether to tell her or not they're having a discussion about whether they should do this or not so there's a contradiction in their conversation that contradicts what they believe what's interesting is that if they do tell her that she has no free will and she believes it she's actually gonna have negative mental effects like it you know be harmful um assuming that she really believes it i don't know if it has that in everyone but you do see studies that show that um so if we do have free will it's a very bad thing to be telling people and the last thing i'll say about that well it's it's interesting because if she's familiar with epistemology and they tell her that she doesn't have free will but she goes okay that's based on a model that has uncertainty and that model may be disproved then she might not have the negative effects of free will so i think everyone should practice being a good bayesian and understand that our models our theories even the one that i just talked about um will have errors and will be shown to be wrong and if reductionists do that they won't call free will belief and free will like comparing them to horoscopes which is just i think a horrible statement and kind of funny that physicists don't like it when like neuroscientists start and biologists start like talking about physics it's kind of strange that physicists are are so sure that we don't have free will when they're not familiar at all with the neuroscience behind agency and free will would you like me to just read a comment in the chat otherwise i think we can kind of come to a close um if there's anything yeah i can do one more question yeah well uh marco and dave and others thanks a lot for the comments and commentary in the chat i hope that bobby i have you check the um live chat replay um and uh sean o'connor wrote thanks for the good and informative talk dr azarian mentioned the idea of evolutionary arms race i was hoping to recommend that he consider if he hasn't how an intra-specific evolutionary arms race may relate i'm not familiar with the specific so uh perhaps like a game theory red queen not between uh two species but within one [Music] yes you would have an evolutionary arms race there too yeah certainly because you're competing with other organisms um uh john maynard smith who i mentioned earlier uh talking about like this causal power of information and biology he uh was the person to bring von neumann's game theory to biological evolution so he did uh he basically showed that there is this you know game theory sort of process being played out i think he his examples use different species but i think within species um you would see the sorts same sort of thing of course if it's not humans you know if you have other

species people are competing organisms are competing you will get increased complexity but there are limits so just because there's these evolutionary arm races uh within and between species there will still be a limit because they have to model the environment and just modeling all these unnecessary variables uh is is wasteful so it will ratchet up complexity in species but i think at least with the planet earth and our biosphere that humans are this sort of leading envelope uh of complexity and um so it really just is open-ended with the most complex species so the most complex species is the one that will evolve toward higher really there might be a limit there there are reasons to think there are limits to how far human biological intelligence can emerge there's i mean can evolve towards there's something called a cephalization limit which is like how big our brain can get because of reasons like having to do with like the skull but what's interesting is part of this process this evolutionary process is popper and newer people like dennett have mentioned continues with science and culture and technology so humans really have open-ended complexity because we're uh we're augmenting ourselves with technology so while biology biological complexity may have limits there's no limits once we start merging with our and we shouldn't see technology as separate from it's an extension of biology so the extended phenotype richard falcons explains like beaver dams considers like the dam that the beaver instinctually builds as part of the whole phenotype so the ecosystem the organism there's really no clear division between the organism and the ecosystem and david chalmers has the extended mind hypothesis which says that our phones are extensions of our minds all of our devices are and for that reason we need to be worried about privacy but we also need to not be scared of technology we just need to use technology to become more human so we're not transcending like humanity and biology technology will allow us to become hyper biological very interesting note perhaps to close it on anytime you want the dot too we can make it happen it can be any format we can invite anyone else on awesome and interesting to think about do you want any last words um if i come back to talk about free one agency it'd be great to have like kevin mitchell or um maybe aircoil who's done a lot of work on uh top-down causation and causal emergence that was really influenced uh influential on the book um i think if you know people you can um twitter tag them or email us and uh email them and cc us or just contact us and we'll do that on our site thanks been super fun thanks daniel really appreciate it great appreciate you reading a draft of the manuscript just like in a few days to give me some feedback that actually helps their little mistakes in the book um i heard you mention marco thanks uh to marco and anybody else who's uh been watching and chiming in with questions really appreciate it sorry for the

information overload maybe watch this again maybe micro dose or um maybe some legal stuff
delta eight or keep stay hydrated